# Auto-Financing Quotes — Secure, Resilient & Event-Driven

Estimated quotes • multi-AZ application • transactional outbox • regional disaster recovery

Users
Customers / dealers

Amazon Route 53

DNS (global)
(routes to healthy region)

DNS (health checks, fails over to Region B if needed)

1. Cognito: customer sign-in + dealer federation
Multi-region user-pool replication

Authentication
(OIDC / SAML)

## REGION A — primary (active)

AWS WAF
WAF rules + HTTPS

AWS ACM

2. Application Load Balancer
(public subnets)

### 3. Quote service — ECS on Fargate

AZ A — ECS task
AZ B — ECS task
AZ C — ECS task

Private subnets • stateless • AZ-loss capacity

### 4. Aurora PostgreSQL

AZ A — Writer (primary)
AZ B — Read replica
AZ C — Read replica

Key tables (in same database)

- quotes (customer, vehicle, terms, status, ...)
- outbox_events (event_id, type, payload, status, ...)

ONE DATABASE TRANSACTION

Also: pricing versions • notification ledger

Aurora storage: six copies across three AZs

5. Outbox publisher on Fargate
Claim committed events • publish • retry

6. SQS Standard

DLQ

7. Email Lambda
Event-ID dedupe; record SES acceptance

8. Amazon SES
Send customer email (verified domain)

Customer email
(quote details)

Read quote / update notification ledger

Email failure does not roll back quote
(duplicate emails remain possible)

**Security (Region A)**

- IAM task roles (least privilege)
- AWS KMS (encryption at rest/in transit)
- AWS Secrets Manager (database credentials, API keys)
- Private subnets (no public DB access)
- VPC endpoints (S3, CloudWatch, ECR, ...)
- Controlled egress (SES, external APIs via NAT / firewall)

## REGION B — hot standby, same multi-AZ stack

Async Global Database replication — quotes + outbox + ledger

Cognito replicated user pool

ALB + ECS on Fargate
(same configuration)

Aurora PostgreSQL
(secondary, read-only)

Publishers & consumers disabled until controlled activation

SQS

Lambda

SES

SQS is regional: replay retained outbox events not confirmed complete; duplicate emails remain possible.

**Operations & DR (both regions)**

- Amazon CloudWatch — Metrics/logs/traces + outbox age / queue age / replica lag
- AWS CloudTrail — AWS API audit → isolated S3 archive
- Terraform — IaC — CI/CD, reviewed plans, protected remote state
- AWS Backup — PITR + isolated cross-region backups

**Illustrative targets:** 99.95% monthly availability | AZ recovery ≤ 2 min | Region RTO 15 min | Saved-quote regional RPO 5 min (validate, not guaranteed)

**Regional failover sequence:**
Fence → promote DB & identity → validate → activate workers → route